

Simulating Wireless Networks in ns-3

ns-3 is a discrete event network simulator that is primarily used in teaching and research. Over the past few issues of OSFY, we have discussed various aspects of ns-3. In this article, we move forward with the simulation of a wireless network.



In the past two issues, we have simulated two wired networks using ns-3. But the world is moving on from wired communication to wireless communication very quickly. The drop in the number of landline telephone connections over the years, summarises the sad story of wired communication. Industry and academia are eagerly waiting for results impacting the wireless communication sector. So it is time for us to move on to simulating wireless networks.

Simulating wireless networks with ns-3

In this section, we will simulate a simple network involving wired and wireless nodes. This is another area where ns-3 clearly outperforms ns-2. In the latter's case, simulating a wired-cum-wireless network is not an easy task. We have to use hierarchical addressing to simulate wired-cum-wireless networks in ns-2. Packet movement in the wireless section of a wired-cum-wireless network is not visible in the NAM animator of ns-2. But in ns-3, we don't have any such problems. We can simulate a pure wireless network involving wireless nodes alone or we can simulate a network involving both wired and wireless nodes. The ns-3 script called *wireless.cc*, which is discussed in this section, will generate an animation trace file called *wireless.xml* and an ASCII

trace file called *wireless.tr*. I have only discussed the code relevant to wireless communication in this article but you can download the complete *wireless.cc* file and all the others mentioned in this article from opensourceforu.ifytimes.com/article_source_code/September2015/ns3.zip.

The following lines of code show the additional header files required for simulating wireless networks. These header files are included in addition to the header files we discussed in the previous example of scripts involving wired networks.

```
#include "ns3/wifi-module.h"
#include "ns3/mobility-module.h"
#include "ns3/internet-module.h"
```

The following lines of code create an object *allNodes* of class *NodeContainer* common to all the other nodes and add two Wi-Fi station nodes to the *allNodes* object. The *NodeContainer* class keeps track of all the node pointers.

```
NodeContainer allNodes;
NodeContainer wifiStaNodes;
wifiStaNodes.Create(2);
allNodes.Add(wifiStaNodes);
```